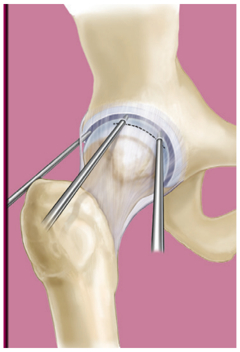
84

Athletic Pubalgia/Core Muscle Injury and Adductor Pathology

Christopher M. Larson, Jeffrey J. Nepple



Athletic pubalgia, or core muscle injury, is an umbrella term describing several anatomic injury patterns present in athletes with groin pain. The term *sports hernia* has fallen out of favor, as it is clearly a misnomer where no true hernia is involved. The correct diagnosis and treatment of this entity can be challenging.[1,2] However, increased recognition and surgical treatment of athletic pubalgia has increased dramatically in the last 20 years. Various terms—including *sports hernia*, *Gilmore's groin*, *osteitis pubis*, *slap-shot gut*, and *sportsman's hernia*—have been utilized to describe the condition.[2-5] There is growing recognition, however, that groin injuries in athletes comprise a complex set of injuries to the musculature of the abdominal wall, adductors, hip joint, pubic symphysis, and sacroiliac joint that can be a source of significant disability.[3,4,6,7] The term *core muscle injury* is also increasingly utilized as an alternative to *athletic pubalgia*. Athletic pubalgia most commonly involves the abdominal wall and adductors but may have significant overlap with conditions of the pubic symphysis (osteitis pubis) and motion-limiting conditions of the hip joint (femoroacetabular impingement).

Although our understanding of the pathophysiology of athletic pubalgia has improved significantly, considerable controversy still remains. Numerous methods of surgical treatment of athletic pubalgia exist, all generally with high rates of return to athletics and short recoveries noted in the literature. Combined treatment of adductor tendon pathology and underlying femoroacetabular impingement also vary. The pathophysiology of athletic pubalgia is most commonly theorized to involve (1) pathology of the rectus abdominis or rectus-adductor aponeurosis, (2) posterior inguinal floor defect, or (3) inguinal or genital neuropathy.[8]

Anatomy

Given the complexity of the anatomy of the hip joint, pelvis, pubic symphysis, and the associated abdominal wall, a thorough understanding of relevant anatomy is important for the clinician assessing an athlete with groin pain. This is particularly true for the anatomic considerations of athletic pubalgia. The pubic symphysis is perhaps the simplest structure relevant to athletic pubalgia and is a nonsynovial amphiarthrodial joint.[9] Static stability of the joint is provided by the disk and four ligaments. The arcuate or inferior ligament has attachments to the inferior articular disk, the inferior attachment of the rectus abdominis, and the adductor and gracilis aponeurosis. The superior ligament spans the space between the pubic tubercles. The anterior ligament blends with fibers of the external oblique and rectus abdominis superficially. The deep portion of the anterior ligament attaches to the intra-articular disk. The posterior ligament is poorly developed. The term *pubic joint* was introduced by Myers and has been used to describe the complex biomechanics of the muscular envelope at the pubis symphysis. The pubic symphysis acts as the fulcrum for forces generated at the anterior pelvis. It represents the common attachment of the confluence of the rectus abdominis fascial sheath with the fascial sheath of the adductor longus; these merge anterior to the pubis to form a common sheath.[9-11] Pathology of other adductors, including the adductor brevis and pectineus, is less commonly seen in athletic pubalgia. Imbalance between the rectus abdominis and adductor muscular and pathology of the common attachment appears to be a major factor in athletic pubalgia.

The abdominal wall has a layered structure. From superficial to deep, the structures of the abdominal wall are skin, fascia, external oblique muscle/fascia, internal oblique muscle/fascia, transversus abdominis muscle/fascia, and transversalis fascia. The posterior fascia is deficient in the lower third of the rectus. Fibers from the rectus, conjoint tendon (fusion of the internal oblique and transversus abdominis fascia), and external oblique merge to form the pubic aponeurosis (also called rectus abdominis/adductor aponeurosis), which is confluent with the adductor and gracilis origin. The conjoint tendon inserts anterior to the rectus abdominis on the pubis (Fig. 84.1).[12]

Weakness of the floor of the posterior inguinal canal may play a role in athletic pubalgia. The inguinal canal is formed anteriorly by the external oblique aponeurosis; posteriorly by the transversalis fascia and conjoint tendon; superiorly by the transversalis fascia, internal oblique, and transversus abdominus; and inferiorly by the inguinal ligament (from the external oblique aponeurosis). The inguinal canal contains the spermatic cord (in males), round ligament (in females), genital branch of the genitofemoral nerve (which supplies motor function to cremaster muscle and sensory function to the scrotum), and ilioinguinal nerve (supplying sensory function to the groin). The role of nerve-mediated pain in athletes with groin pain remains somewhat poorly understood and controversial. Knowledge of the nerve anatomy of the groin is important in elucidating the potential role of each nerve in athletic pubalgia. One possible source of pain has been theorized to be the result of entrapment of the genital branch of the genitofemoral or ilioinguinal nerve.[13] Weakness of the posterior inguinal floor has been proposed to result in dynamic compression of the

Abstract

Athletic pubalgia, or core muscle injury, is an umbrella term describing several anatomic injury patterns present in athletes with groin pain. Various terms, including *sports hernia, Gilmore's groin, osteitis pubis, slap-shot gut,* and *sportsman's hernia,* have been utilized to describe the condition. There is growing recognition, however, that groin injuries in athletes comprise a complex set of injuries to the musculature of the abdominal wall, adductors, hip joint, pubic symphysis, and sacroiliac joint that can be a source of significant disability. The term *core muscle injury* is also increasingly utilized as an alternative to *athletic pubalgia.* Athletic pubalgia most commonly involves the abdominal wall and adductors but may have significant overlap with conditions of the pubic symphysis (osteitis pubis) and motion-limiting conditions of the hip joing (femoroacetabular impingement). Although our understanding of the pathophysiology of athletic pubalgia has improved significantly, considerable controversy still remains. Numerous methods of surgical treatment of athletic pubalgia exist, all generally with high rates of return to athletics and short recoveries noted in the literature.

Keywords

athletic pubalgia
sports hernia
core muscle injury

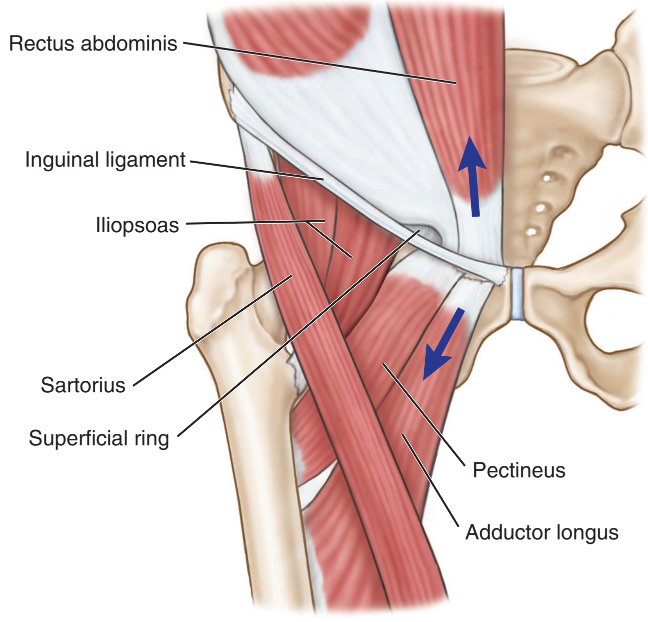


Fig. 84.1 Injury to the abdominal wall at the fascial attachments of the rectus and adductors onto the pubis is implicated in athletic pubalgia. (From Omar IM, Zoga AC, Kavanagh EC, et al. Athletic pubalgia and "sports hernia": optimal MR imaging technique and findings. *Radiographics.* 2008;28[5]:1415–1438.)

genital branch of genitofemoral nerve. The symphysis itself is innervated by branches of the pudendal and genitofemoral nerves.[9]

Historical Background

Groin injuries were discussed in the medical literature as early as 1932, when Spinelli reported on pubic pain in fencers.[16] Nearly 50 years later, in 1980, Gilmore recognized a "severe musculotendinous injury of the groin" in three professional soccer players. He identified a triad of pathology including injuries to the external oblique aponeurosis and conjoint tendon, avulsion of the conjoint tendon from the pubic tubercle, and dehiscence of the conjoint tendon from the inguinal ligament.[3] He later reported a 97% return to sport after surgical repair.[17]

In 1993, Hackney first used the term *sports hernia* to describe a syndrome of groin pain in athletes that had failed nonsurgical management. At surgery, he identified weakening of the transversalis fascia with separation of that fascia from the conjoint tendon, dilation of the inguinal ring, and one case of a small direct hernia. He treated all patients with a surgical repair of the posterior inguinal wall and obtained an 87% return to sport in 15 athletes.[18] Irshad et al. described "hockey groin syndrome" in 22 National Hockey League players in 2001.[19] They found tearing of the external oblique aponeurosis and entrapment of the ilioinguinal nerve. Meyers proposed that use of the term *athletic pubalgia*[4] would be more appropriate for the constellation of injuries to the abdominal wall, adductors, and pubic symphysis presenting with pubic area or inguinal pain than the more commonly used *sports hernia*.[2] He proposed that the primary pathology in athletic pubalgia was an imbalance between the strong adductors and the relatively weak abdominal muscles. More recently, the term *core muscle injury* has also been increasingly used to describe athletic pubalgia. Meyers describes 17 different variants of athletic pubalgia, the most common of which were multiple tears or detachment of the anterior and anterolateral fibers of the rectus abdominis from the pubis and combined injuries to the rectus and adductors.[20]

Over the last two decades, treatment of athletic pubalgia has become increasingly common in high-level athletes involved in cutting/pivoting and acceleration/deceleration sports, including soccer, football, and ice hockey. Between 2012 and 2015, some 4.2% of athletes of the National Football League (NFL) had undergone surgical treatment of athletic pubalgia (most commonly defensive backs and wide receivers).[21]

As the understanding of intra-articular hip pathology has improved, there has been increasing recognition that acetabular labral pathology and femoroacetabular impingement can coexist with athletic pubalgia.[7,10] Significant overlap between athletes with athletic pubalgia and femoroacetabular impingement is seen in clinical practice but has only recently been recognized.[7,10] Feeley et al. described the "sports hip triad" of labral tears, rectus abdominis strain, and adductor strain.[22] The loss of internal rotation of the hip appears to play a role in the development of groin pain and osteitis pubis.[14] Based on these observations, Birmingham et al. performed a cadaveric study looking at the effect of femoroacetabular impingement on rotation at the pubic symphysis in a cadaveric model.[15] At higher torque values, motion through the symphysis was much greater in cadavers with simulated cam morphology than in the native hip state. This supports the previous hypothesis that the altered rotational profile seen in the setting of femoroacetabular impingement contributes to altered mechanics at the pubic symphysis through stress transfer. Limitations in range of motion (ROM) at one location can result in additional motion occurring at other locations.

Larson et al. reported the outcomes of surgical treatment in a subset of athletes with coexistent femoroacetabular impingement and athletic pubalgia.[7] Failure to treat both pathologies resulted in a low return to sport (25% if athletic pubalgia was addressed in isolation, and 50% if the intra-articular hip pathology was addressed in isolation). This prompted the development of a surgical protocol to address both etiologies under the same anesthesia when athletes present with both symptomatic athletic pubalgia and intra-articular hip disorders (FAI). Using this approach, these investigators achieved an 85% to 93% return to sport.[7]

In addressing the athlete with groin pain, it is therefore important to carefully investigate other potential sources of the athlete's pain. Intra-articular hip pathology and associated hip pathomorphology must be carefully screened for if symptomatic and limiting. This assessment should also take into account the high rates of potential asymptomatic labral pathology on magnetic resonance imaging (MRI) in this population. Additionally, it is important to be aware of the broader differential diagnosis of groin pain. Disorders of the gastrointestinal, genitourinary, or gynecologic systems must also be considered, with several authors identifying tumors, Crohn disease, and other pathologies in the evaluation of the athlete with groin pain.[2,21]

HISTORY

Athletic pubalgia is particularly common among those who participate in sports requiring repetitive twisting, pivoting, and cutting motions, as well as activity requiring frequent acceleration and deceleration. Trunk hyperextension and hip hyperabduction place significant stress across the pelvis, including the rectus abdominis and adductors. Soccer, football, ice hockey, and rugby have a particularly high incidence, with a lower incidence reported in basketball and baseball.[3,19,21] Meyers noted that in the 1980s, less than 1% of his patients with athletic pubalgia were female. Over the last two decades, however, that has changed dramatically, and female patients now represent 15% of patients presenting with athletic pubalgia.[20]

Athletes with athletic pubalgia may report an insidious onset of groin pain or less commonly have a more acute presentation. Most athletes report pain with athletic activities but not at rest. Although there is some variability in the location and characteristic of symptoms, Meyers et al.[6] found that all of their athletes reported lower abdominal pain with exertion, whereas 92% had minimal to no pain at rest. Forty-three percent developed bilateral symptoms and 67% developed adductor pain after the onset of lower abdominal pain. Pain may also radiate to the rectus, perineum, or testicular region.[7] Exacerbation typically occurs during activities involving kicking, acceleration, and pivoting.[6,22] Abdominal crunches or situps, coughing, or sneezing may also reproduce symptoms. Osteitis pubis presents in a similar fashion, with pubic pain that may radiate to the adductors and is typically exacerbated by weight bearing.[21] Intra-articular hip pathology from labral tears or femoroacetabular impingement typically presents with deep anterior groin pain that may radiate to the anterior thigh and lateral hip. There is considerable overlap between intra-articular hip symptoms and athletic pubalgia symptoms, which, when combined with high rates of potentially asymptomatic pathology of radiographs and MRI, increases the complexity of making an accurate diagnosis in this challenging patient population.

PHYSICAL EXAMINATION

Assessment for athletic pubalgia should begin with palpation of the pubic symphysis, insertion of the rectus abdominis, adductor origin, external and internal obliques, transverses abdominis, pectineus, gracilis and inguinal ring for areas of tenderness. Provocative testing can be useful to reproduce symptoms, including simulated coughing, Valsalva maneuver, resisted situps (46%), (Fig. 84.2), or resisted hip adduction.[6] With traditional assessment for inguinal hernia, there is no palpable hernia, but this should be ruled out. There is usually tenderness around the conjoint tendon, pubic tubercle (22%), adductor longus (36%), superficial inguinal ring, or posterior inguinal canal.[6,28,29] Detailed assessment of hip ROM including provocative testing is warranted given the interplay between athletic pubalgia and FAI.

Exam findings for osteitis pubis frequently overlap (or coexist) with athletic pubalgia and include tenderness of the pubic symphysis (67%), adductor origin tenderness (59%), pain with the adductor squeeze test (96%),[5] and apprehension throughout the hip's ROM.[30] More severe cases of osteitis pubis may present with a classic waddling gait pattern.[28]

Fig. 84.2 Pubic and/or groin pain with resisted situps is associated with athletic pubalgia. (From Minnich JM, Hanks JB, Muschaweck U, et al. Sports hernia: diagnosis and treatment highlighting a minimal repair surgical technique. *Am J Sports Med.* 2011;39[6]:1341–1349.)

Diagnostic Injections

As indicated earlier, the symptoms of femoroacetabular impingement, athletic pubalgia, osteitis pubis, and adductor strain can present with overlapping symptomatology and physical examination findings. An intra-articular injection of local anesthetic into the hip followed by physical examination or by having the athlete perform activities that typically provoke the pain can be useful in the evaluation of this population as well as in the resultant decision-making. Pain that resolves or significantly improves with this injection can then be assumed to be related to intra-articular hip pathology and be treated accordingly.[7] Persistent pain the lower abdominal and proximal adductor regions after intra-articular injections is consistent with coexistent athletic pubalgia. Similarly, injection of the pubic symphysis may be useful for confirming the diagnosis of osteitis pubis. Radiocontrast dye can be used for the symphyseal injection, and extravasation of this dye up the rectus abdominis tract or down the adductor tract may indicate underlying pathology consistent with athletic pubalgia. Adductor pathology can be ruled in or out with an injection of anesthetic into the pubic cleft. Diagnostic injections can be used in the assessment of other pathology around the hip and groin as well. If psoas disorders are suspected, a diagnostic psoas bursal injection with anesthetic can be carried out. If impingement is suspected from the anterior inferior iliac spine, a subspinal injection can be carried out. Together these targeted diagnostic injections can be useful adjuncts to clinical assessment and aid in treatment decision-making, particularly in decisions regarding combined surgical treatments.

IMAGING

Radiographic Analysis

Plain radiographs are vital in the initial evaluation of the athlete with hip or groin pain. A number of pathologies including osteitis pubis, avulsion fractures, stress fractures, apophysitis, osteoarthritis, femoroacetabular impingement, and hip dysplasia may be identified on radiographs. It is essential to obtain good-quality, properly oriented images according to an established imaging protocol.[32]

The anteroposterior (AP) pelvic view (Fig. 84.3) may be used to evaluate the pubic symphysis for evidence of osteitis pubis, including sclerosis, fragmentation, and cyst formation within the pubic ramus as well as symphyseal widening. Most authors now favor performing this radiograph in the standing position to best simulate the functional position of the pelvis, which commonly differs from the supine position. In evaluating for femoroacetabular impingement, also assed are deformities of the femoral head neck and acetabular depth and version. In the adolescent athlete, the AP view can be useful to identify apophyseal avulsion injuries. Additionally, stress fractures of the femoral neck and pubic rami and sacroiliitis may be identified.[12]

In cases of possible pubic symphyseal instability associated with osteitis pubis, stress radiographs may be useful. The stability of the pubic symphysis can be determined on single-leg-stance AP views (the "flamingo view"). Symphyseal widening greater than 7 mm or vertical translation greater than 2 mm on a single-leg-stance view suggests instability of the pubic symphysis.[30]

Magnetic Resonance Imaging

MR arthrography has been used for the assessment of intra-articular hip pathology. More recently, noncontrast 3.0 Tesla MR has been evaluated and found to be sensitive for intra-articular pathology including articular cartilage injury and labral pathology. The use of arthrography on MR is now dependent on the surgeon and institutional preference. Routine assessment of common sites of pathology in athletic pubalgia on hip MRI can be useful in athletes with groin pain, but dedicated protocols for pelvic MRI are optimal for visualization. Coronal oblique and axial oblique sequences through the rectus insertion and pubic symphysis should be obtained in addition to standard sagittal, coronal, and axial sequences. MRI is 68% sensitive and 100% specific for rectus abdominis pathology as compared with findings at surgery (the gold standard) and is 86% sensitive and 89% specific for adductor pathology. MRI is 100% sensitive for osteitis pubis.[33] Nonarthrographic studies may be preferred for in-season athletes to avoid the potential for irritation secondary to the administration of intra-articular contrast.

The MRI should be evaluated in a systematic fashion for pathology consistent with athletic pubalgia. The pubic bones should be evaluated for edema, subchondral sclerosis, and cysts suggestive of osteitis pubis (Fig. 84.4).[11,34] Evaluation of the tendinous insertions of the core muscles around the pubic symphysis should then be performed (Fig. 84.5). Frequent findings include fluid signal within the rectus abdominis or adductor origin, thickening of either structure, peritendinous fluid, or partial or complete disruption of either tendon. Most commonly there is a confluent fluid signal extending from the anteroinferior insertion of the rectus abdominis into the adductor origin, with corresponding fluid signal in the pubis.[33]

Dynamic Ultrasonography

Dynamic ultrasound has become a common form of diagnostic evaluation in patients with athletic pubalgia. Ultrasound takes advantage of the dynamic nature of the pathology in athletic pubalgia. Visualization of the structures muscular insertions at the pubis and the posterior inguinal canal can be performed at rest and with stress. A convex anterior bulge of the posterior inguinal canal is a common finding in the setting of posterior

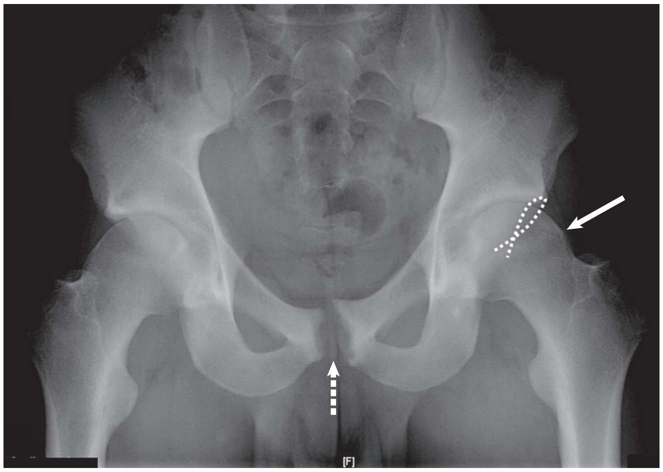


Fig. 84.3 Well-aligned anteroposterior radiograph of the pelvis. Note the coccyx centered over the pubic symphysis and a centimeter proximal. On this radiograph, there are finding consistent with osteitis pubis *(large arrow)*, including erosion, irregularity, and cyst formation. Additionally, there is a positive crossover sign, suggesting possible pincer impingement and a large cam lesion on the femoral head-neck junction *(small arrow)*.

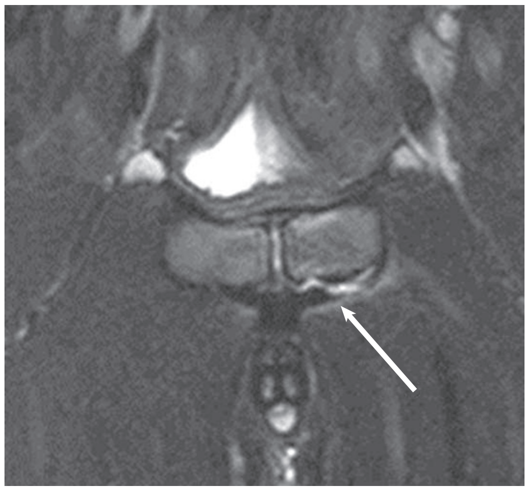


Fig. 84.4 Oblique axial fat-suppression T2 magnetic resonance image of the pubic symphysis. There is disruption of the left rectus aponeurosis as it inserts on the anterior aspect of the superior pubic ramus *(tip of the white arrow)*.

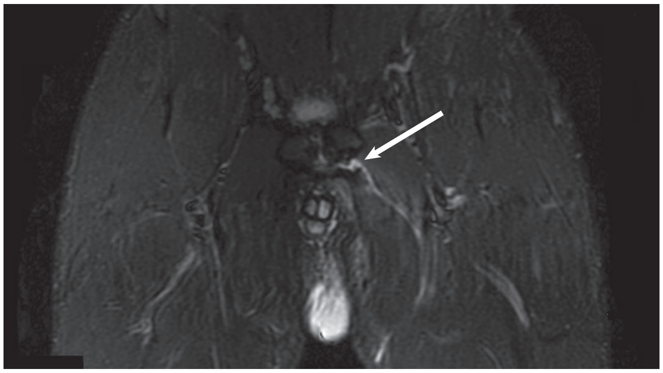


Fig. 84.5 Coronal fat-suppression T2 image of the pelvis demonstrating injury to the left adductor longus at its origin on the anteroinferior aspect of the superior pubic ramus. There is a fluid signal coursing linearly distally through the proximal muscle fibers *(tip of the white arrow).*

inguinal canal deficiency.[35] Ultrasound is dependent on the expertise and experience of the ultrasonographer and is most applicable to high-volume centers.

DECISION-MAKING PRINCIPLES

Treatment decisions must take into account a number of factors including the degree of limitation and ability to participate in the athlete's respective sport, duration of symptoms, pathology identified on physical examination and imaging, response to prior treatment modalities, and where the athlete is with respect to his or her training, sport season, and upcoming athletic events. The common locations of pain are the groin (FAI, athletic pubalgia, adductor), lower abdomen or pubic symphysis (athletic pubalgia, adductor longus), posterior hip (FAI, proximal hamstring, low back, sacroiliac joint, sciatic nerve entrapment disorders), or lateral thigh/hip (iliotibial band or gluteus medius/minimus). Conservative management should be attempted prior to surgical intervention. If no previous treatment has been attempted at presentation, a conservative trial should include rest, nonsteroidal antiinflammatory drugs (NSAIDs), physical therapy, and injections in select situations. Physical therapy is focused on restoring core muscle strength and correcting any underlying imbalance between core muscle groups. If the athlete continues to be symptomatic after 6 to 12 weeks of nonsurgical treatment, then surgery may be considered. The timing of the surgery depends on the degree of disability and the time point in the season. If the athlete must participate and he or she is productive and functional, a delay until after the season may be attempted. Surgery can then be performed at the end of the season if the athlete remains symptomatic. If the athlete is not able to compete at a reasonable level, then in-season or season-ending surgery can be considered. If there is a combined pathology such as FAI and athletic pubalgia, these can be surgically addressed at the same setting or staged in order to minimize postoperative rehabilitation time, total time lost from participation, and potential need for subsequent surgery. There is, however, no evidence to show that carrying out these procedures separately or in a staged manner, has any negative impact on outcome.[7]

TREATMENT OPTIONS

Adductor Strain

Adductor-related pain can present as an acute injury or a chronic condition. Muscle imbalance between the abductors and adductors does appear to contribute to injury of the adductors. Tyler et al. found that professional hockey players were 17 times more likely to sustain an adductor strain if their adductor strength was less than 80% of their abductor strength.[36] In a follow-up study, they were able to demonstrate a clinically and statistically significant decrease in adductor strains in the same population with institution of a preventative adductor strengthening program.[37] Recognition of at-risk athletes with such muscular imbalance thus may play an important role in injury prevention in this population.

Nonoperative Management

Treatment starts with a brief period of rest, judicious use of ice and NSAIDs, and the institution of core and lower extremity strengthening programs. Once the athlete is able to perform a pain-free concentric contraction of the adductor against resistance, the program can be progressed to core strengthening and adductor-specific exercises. Identification and treatment of any deficits or imbalances in the contralateral extremity is useful at this stage as well. It has been suggested the athlete can progress to sport-specific training when the adductor strength is 75% of the ipsilateral abductor and passive ROM is normal.[37,38]

One randomized clinical trial is available; it presents an 8- to 12-year follow-up after nonoperative management of adductor-related groin pain.[39] The initial study randomized 59 athletes to either (1) a passive rehabilitation protocol consisting of modalities and stretching or (2) an active rehabilitation protocol emphasizing strengthening of the core, abductors, and adductors.[39] In the initial study period, 23 of 29 patients treated with 8 to 12 weeks of active therapy had returned to sport by 4 months after initiating treatment.[38] In the passive therapy group, only 4 of 30 patients had returned to sport at the 4-month mark. At 8 to 12 years after treatment, 50% of the active rehabilitation group had no adductor pain with activity, no groin pain during or after activity, and were active in athletic activity at or one level below their previous level of athletic activity in the same sport, whereas only 22% of the passive therapy group met the same criteria.

The evidence for platelet-rich plasma (PRP) injections about the hip and pelvis, however, is lacking. There is one case report of injecting a complete tear of the adductor longus with PRP followed by a return to competitive soccer without surgery.[40] However, successful nonoperative treatment of this injury has been reported without PRP as well. Dallaudiere et al.[41] reported the clinical and ultrasound-based outcomes of 40 patients with chronic hip pain due to either the adductor or hamstring tendons. Improvements in WOMAC scores and decreased lesion size were noted at mean follow-up of 20 months.

Operative Management

Acute adductor injuries generally do not require surgical treatment. Schlegel et al. reported on 19 professional football players who sustained spontaneous rupture of the adductor longus. Twelve of 19 had groin or abdominal pain that preceded acute rupture. Of the 19 athletes identified, 14 were treated conservatively and 9 underwent surgical repair with suture anchors. The nonoperative group returned to play at an average of 6 weeks after injury with no noted strength deficits. The operative group returned to play at an average of 12 weeks after injury, and 20% experienced wound complications.[42] Given that there are minimal functional deficits after spontaneous rupture of the adductor at its origin, controlled release in the context of a surgical procedure might similarly have few long-term functional consequences.

Surgical treatment of chronic proximal adductor pain is indicated when an athlete has failed 3 to 6 months of conservative management. Successful outcomes of a variety of techniques—including complete proximal release, percutaneous partial release, or fractional lengthening—has been reported, although the degree of residual weakness may differ. Akermark and Johansson published a case series of 16 competitive athletes with long-standing (mean 18 months) and recurrent groin pain localized on physical exam to the adductor origin.[43] All athletes had failed conservative management with rest, stretching, NSAIDs, and corticosteroid injections. Surgical treatment involved open tenotomy 1 cm from the adductor origin. All patients were improved and able to resume sporting activities at some level. Ten of 16 patients were pain-free. The operated leg was weaker in adduction torque in all patients after full recovery, although not necessarily symptomatic. Atkinson et al. reported on 48 athletes who underwent 68 percutaneous adductor tenotomies for adductor strain. All patients were significantly restricted in their chosen sport preoperatively, and 54% were able to return to full sport at a mean of 18.5 weeks.[44] Proximal adductor release does appear to result in some persistent adductor weakness, although this has not been demonstrated to affect return to sport.[45]

Given the overlapping symptoms of core muscle injury, including adductor-related pain and athletic pubalgia, controversy exists on the optimal treatment or treatments. In athletes undergoing athletic pubalgia surgery, the combination of treatment of proximal adductor pathology in the presence of significant adductor symptoms may play a role in optimizing outcome. Some authors have also suggested that the treatment of adductor pathology may improve athletic pubalgia symptoms even in the absence of surgical treatment.

Athletic Pubalgia/Sports Hernia

Nonoperative Management

Generally a brief period of rest is indicated. Physical therapy should emphasize core strengthening and the identification and treatment of weakness and restricted motion in the musculature of the hip and pelvis. Ice and NSAIDs can be helpful for managing pain. Avoidance of heavy weight, deep hip flexion, low repetition, squats/lunges/cleans is advisable during this period of rehabilitation. As with adductor pathology, the role of PRP or other biologics remains to be defined. Case reports indicate that injection into the distal rectus insertion can have a good short-term outcome.

Operative Management

A variety of surgical procedures have been described in the literature for the treatment of athletic pubalgia, all with generally excellent reported outcomes. These procedures can be classified as involving primary repair, mesh-based repair, or minimal repair. Gilmore first described “Gilmore’s groin” as a cause of chronic groin pain in athletes in 1980.[11] His repair involves plication of the transversalis fascia, reapproximation of the conjoint tendon to the inguinal ligament, and approximation of the external oblique aponeurosis. Utilizing this technique, the reported return to sport has been 96% to 97% within 10 to 12 weeks.[16,44] Hackney published his technique in 1993. On surgical exploration, weakness of the transversalis fascia with separation from the conjoint tendon and dilatation of the internal inguinal ring was identified. A direct repair was performed with the goal of reconstitution of the internal inguinal ring, plication of the transversalis fascia, and apposition of the conjoint tendon and internal inguinal ring.[17] Hackney presented a series of 15 patients with an average duration of groin pain for 20 months prior to surgical intervention. Postoperatively, swimming and cycling were allowed at 3 weeks, running at 4 to 5 weeks, and sport-specific exercise at 6 weeks. Eighty-seven percent of the athletes were able to resume full participation in sporting activities.

Meyers has published the largest cohorts on the treatment of athletic pubalgia utilizing a “pelvic floor repair.”[6] This involves an open surgical approach with reattachment of the anteroinferior rectus abdominis to the pubis and a variation of an adductor release. In 276 athletes evaluated for groin pain, 157 had clinical symptoms and exam findings consistent with athletic pubalgia, subsequently undergoing primary pelvic floor repair. Imaging and intraoperative findings were somewhat variable and included unilateral or bilateral tearing or scarring of the rectus, tearing of the external oblique aponeurosis, and scarring of the adductor origin. Postoperatively, 152 were able to return to their preinjury levels of competition. In 2008, Myers reported on his experience of over 20 years and 5460 surgeries in a population that included 83% athletes (most commonly soccer, football, and ice hockey). He noted the evolving understanding of the pathophysiology of athletic pubalgia and associated treatments aimed at correcting the underlying pathology. This included performance of 26 different procedures and 121 combinations of these in the group. Overall, 95.3% of athletes were able to return to sports by 3 months postoperatively. Myers reported that athletes operated on more recently were following either a 3- or 6-week return-to-sport protocol, after which the majority were able to return to full participation.[18] Management of the genital branch of the genitofemoral nerve and ilioinguinal nerve remains controversial, with some surgeons advocating routine resection whereas others advise against this.

More recently a laparoscopic approach has been reported with the potential of allowing a more rapid recovery. Genitsaris et al. identified 131 athletes with groin pain who had failed 2 to 8 months of physical therapy. All patients underwent laparoscopic bilateral mesh repairs (transabdominal preperitoneal), with the mesh extending from the pubis to the anterosuperior iliac spine bilaterally. The peritoneum was closed over the mesh. All patients were able to return to full sports participation at 2 to 3 weeks postoperatively.[47]

Kluin et al.[48] identified 14 athletes with persistent groin pain despite 3 months of conservative management. During

Authors' Preferred Treatment

The authors prefer a fractional lengthening or recession of the adductor longus performed through a 2-cm incision (Fig. 84.6). The adductor tendon is released 3 to 5 cm distal to its origin, leaving the underlying muscle intact (Fig. 84.7). This may have less potential for longer-term adductor weakness. In the setting of significant degenerative tearing and tendinosis proximally, however, a proximal release near the origin is performed.

Fig. 84.6 Skin markings for the 2-cm incision utilized for open adductor tenotomy. Note the *dotted lines* marking the borders of the adductor tendon and the location of the incision 4 to 5 cm distal to the origin.

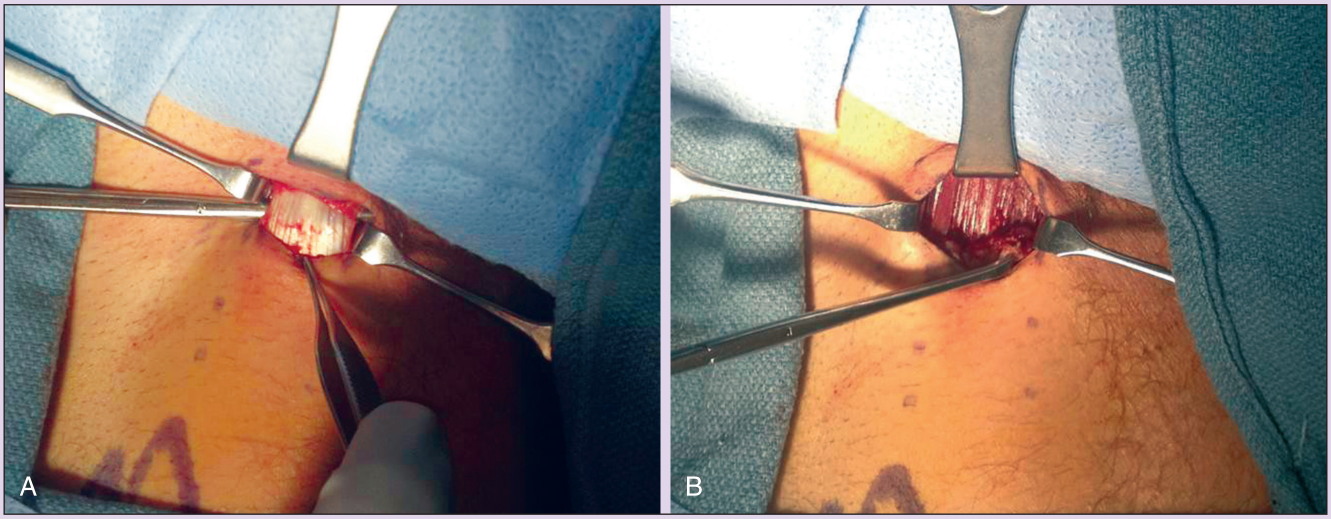


Fig. 84.7 (A) Intraoperative image of the exposed adductor longus tendon. (B) Intraoperative image of the underlying adductor muscle belly after release of the tendinous portion.

laparoscopic exploration, 9 occult inguinal hernias, 4 occult femoral hernias, and 3 preperitoneal lipomas were identified, among other pathology. Treatment was with laparoscopic tension-free mesh repair. Overall, 13 of 14 athletes were able to return to sport by 3 months. Ingoldby et al. compared traditional open and laparoscopic results in 28 athletes. Fourteen open procedures and 14 laparoscopic procedures were performed. Full training was resumed at an average of 5 weeks after the open procedures and 3 weeks after the laparoscopic procedures.[49]

Muschaweck and Burger developed a "minimal repair" technique in 2003 (Fig. 84.8).[50] They identified patients with athletic pubalgia on the basis of clinical history of groin pain radiating to the inner thigh, pubis, testicles, or scrotum. Findings on examination included tenderness on palpation of the internal inguinal

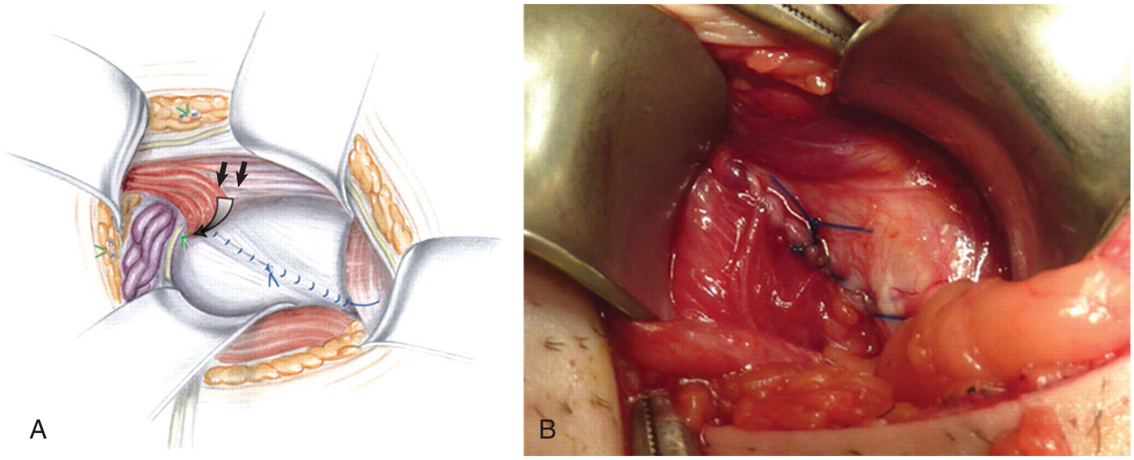


Fig. 84.8 (A) Illustration and (B) intraoperative image of tension-free repair of the defect in the posterior wall of the inguinal canal. (From Minnich JM, Hanks JB, Muschaweck U, et al. Sports hernia: diagnosis and treatment highlighting a minimal repair surgical technique. *Am J Sports Med.* 2011;39[6]:1341–1349.)

ring; ultrasound confirmed bulging of the posterior inguinal wall with Valsalva. The "minimal repair" that was subsequently implemented involved decompression (if intact) or resection (if damaged) of the genital branch of the genitofemoral nerve with a tension-free suture repair of any defect in the posterior wall of the inguinal canal. this procedure can be performed without general anesthesia. Return to sport generally occurred by day 14. A total of 132 such procedures were performed in 128 patients including 89 high-level athletes. Overall, 83.7% of the athletes were fully competitive in their sport at 28 days postoperatively. In a more recent cohort of 129 prospectively enrolled patients, similar outcomes were demonstrated with return to training at 7 days postoperatively.[51]

Paajanan et al. performed a prospective randomized controlled trial comparing nonoperative management to laparoscopic sports hernia repair. All athletes reported 3 to 6 months of symptoms prior to enrollment in the study. These patients were either professionals or high-level recreational athletes. The group randomized to nonoperative management underwent a period of rest followed by physical therapy, local injections of corticosteroids, and use of oral NSAIDs for a total of 8 weeks. At 1 month, 20% of athletes had returned to sport; by 3 months, 27% had returned to sport. At 6 months, 7 of 30 had elected to switch to the operative arm of the study. At 1 year, 15 had returned to sport and 14 reported complete relief of pain. Operative management consisted of a laparoscopic procedure with insertion of a preperitoneal mesh repair. Adductor tenotomy was performed in six patients who had adductor tenderness preoperatively. Of these, 67% had returned to sport at 1 month postoperatively and 90% had returned at 3 months. Twenty-nine of 30 athletes had returned to full participation and were pain-free at 1 year after surgery. Pain scores and patient satisfaction scores were better in the operative group at all points up to a year out from surgery.[52]

Osteitis Pubis

Nonoperative Management

Options for nonoperative management of osteitis include rest, use of therapeutic modalities, NSAIDs and corticosteroid injections, and rehabilitation focusing on strengthening of the core and pelvic musculature. Currently all available studies of nonoperative management of osteitis pubis provide Level IV evidence.[53]

In a prospective cohort study, Verrall et al. identified 27 professional Australian-rules football players with chronic groin injuries diagnosed on history and physical exam. On MRI, these athletes had findings consistent with osteitis pubis. Treatment consisted of 12 weeks of non-weight bearing. Swimming and upper-body lifting activities were allowed initially. Cycling was allowed at 3 weeks, along with a core/pelvic strengthening program as long as this did not provoke pain. Stair stepping was initiated at 6 weeks, followed by a graduated running program at 12 weeks. With this protocol, 89% were able to return to sport by 1 year after initiation of treatment. By the second season, 100% of the athletes had returned and 81% were without symptoms.[13]

Rodriguez et al.[54] identified 44 professional soccer players with osteitis pubis on the basis of history, physical exam, radiographic findings including irregularity of the pubic symphysis and rami, and bone scan demonstrating increased uptake in the pubic rami. Thirty-five athletes underwent a conservative rehabilitation protocol that included multiple therapeutic modalities (electrical stimulation or ultrasound, ice massage, and infrared laser) for the first 14 days in conjunction with high-dose NSAIDs. Subsequently the athletes followed a gradual activity progression including stretching; strengthening of the adductors, abductors and abdominal muscles; running progression; plyometrics; and

Authors' Preferred Treatment

The authors refer patients with athletic pubalgia to an experienced general surgeon with an interest in sports hip injuries. Ideally the treatment performed should be athlete-specific and should address the pathology present, which would include oblique and transverses abdominis injury and tears, rectus abdominis and adductor aponeurosis disruptions, and proximal adductor–related injury and pathology. The current literature does not demonstrate evidence of superiority of primary repairs, mesh reinforcement, minimal incision, laparoscopic repairs, or broad pelvic floor repairs.

kicking progression. Athletes with mild symptoms returned to sport at an average of 3.8 weeks after initiation of treatment. Those with moderate symptoms returned to sport 6.7 weeks after initiation of treatment. No long-term follow-up was available.

There is some Level IV evidence in the literature to support the use of additional modalities including injection of corticosteroids into the pubic symphysis. O'Connell et al. found that 14 of 16 in-season athletes were able to return to sport within 48 hours of symphyseal injection of corticosteroid with significant pain relief. However, at 6 months, only 5 of 16 were pain-free, and 4 had required an additional period of rest before returning to full activity.[55]

Holt et al.[56] presented a case-control study of 12 athletes at the University of Wisconsin presenting with physical examination and radiographic findings consistent with osteitis pubis. In the initial arm, 9 athletes underwent 4 months of conservative management consisting of rest, NSAIDs, stretching, and a gradual return to sport-specific activities. After failure of this protocol to resolve symptoms, 8 athletes underwent symphyseal injection of corticosteroids. Of these, 7 were able to return to sport after one to three injections and an additional period of rest ranging from 4 to 24 weeks. In the second arm, three athletes with osteitis pubis were identified; they underwent 7 to 10 days of conservative management, and after failure of early treatment, underwent injection. All three were able to return to sport within 2 weeks and remained symptom-free at 1 year after injection. The authors advocated early consideration of corticosteroid injection after diagnosis of osteitis pubis, as it may allow early return to sport and sustained symptom relief.

Operative Treatment

Operative treatment of osteitis pubis is generally considered a salvage technique and used most frequently in the scenario of demonstrable instability on radiographic studies or failure of several months of conservative management. Techniques described in the literature include curettage of the symphysis,[57] wedge resection,[58] mesh reinforcement of the symphysis, and arthrodesis of the symphysis with bone graft and compression plating.[29]

Wedge resection of the pubic symphysis was first described by Schnute in 1961[59] in a case series. He reported that use of the wedge resection resulted in significant clinical improvement patients who were severely limited by pain. Grace et al.[58] reported on 10 patients who underwent wedge resection at an average of 32 months after onset of symptoms. Of these, 3 patients developed chronic pain and another required fusion of the sacroiliac joints for posterior instability that developed secondary to the wedge resection. Moore et al. reported two additional patients who developed sacroiliac instability after wedge resection.[60] No published study documents the use of this procedure in high-level athletes, and given the relatively high risk of developing late posterior sacroiliac instability, this technique should be used cautiously in the athletic population.

Williams et al. described fusion of the pubic symphysis with compression plating and bone grafting as a viable treatment option for chronic osteitis pubis.[30] They treated seven rugby players with chronic groin pain diagnosed as osteitis pubis on physical examination and radiographic findings. Surgical treatment involved excision of the articular surface of the pubis, cancellous bone grafting, and application of a compression plate. One athlete also underwent mesh repair by Gilmore after injuries to the external oblique aponeurosis and conjoint tendon were identified. Postoperatively the athletes returned to sport at around the 6 months. All seven athletes were able to return to sport and at an average of 52 months after surgery, all were pain-free.

In the largest published study of operative treatment of osteitis pubis in athletes, Radic et al. treated 23 patients with curettage of the pubic symphysis.[57] Surgical treatment involved curettage of the articular surface until all articular cartilage was removed and bleeding bone was encountered. No stabilization of the joint was performed. Initially, 70% were able to return to their previous level of activity at an average of 5.6 months after surgery. However, with longer follow-up, 39% remained asymptomatic and 26% experienced a one-time recurrence of their symptoms that resolved with rest. Yet 30% of their cohort were never able to resume their preinjury level of activity and did not consider the surgical procedure to have been worthwhile. Mesh reinforcement of the pubic symphysis was described by Paajanen et al. in 2005.[61,62] They identified 16 athletes with clinical histories suggestive of osteitis pubis and MRI demonstrating pubic bone marrow edema and a positive bone scan. Eight patients with more severe symptoms, who failed 6 months of conservative management, elected to undergo surgical intervention, which involved laparoscopic placement of mesh behind the pubis held in place with titanium tacks. Two athletes had concomitant adductor or gracilis release. Seven of eight operatively treated patients returned to sport at an average of 2 months postoperatively. At an average of 2.7 years after surgery, all surgically treated patients were competing at their preinjury levels of athletics. The authors also reported on eight patients with more mild symptoms who were treated nonoperatively. Four of eight conservatively managed patients were able to return to their preinjury level of activity after 1 to 1.5 years of conservative management and four were competing with some pain at one level below their preinjury level.[55] The authors felt that surgical management with mesh repair allowed their cohort to return more quickly to sport with better resolution of pain than with conservative management.

There has been one report of treatment of osteitis pubis with an endoscopic decompression of the pubic symphysis in a chronic case that occurred in association with FAI.[63] The FAI was also treated arthroscopically at the same sitting and the patient was improved with resolution of the presenting waddling gait. Matsuda et al.[64] reported a combined case series of seven patients treated with endoscopic pubic symphysectomy combined with arthroscopic treatment of FAI. Two patients demonstrated postoperative scrotal swelling, one to the size of a volleyball, which resolved with observation. Good outcomes were reported in five of seven patients without evidence of instability, whereas one patient required fusion due to persistent symptoms. Further studies are needed to clarify the role of endoscopic surgery for osteitis pubis.

Meyers initially excluded athletes with osteitis pubis from surgical treatment of athletic pubalgia.[6] However, as clinical experience improved, prior patients who had been diagnosed with osteitis pubis, in addition to athletic pubalgia, were reviewed after undergoing isolated surgical treatment of athletic pubalgia. Similar outcomes were seen in patients with and without adductor release. This reinforces the hypothesis that athletic pubalgia and osteitis pubis represent a spectrum of injuries caused by abnormal forces around the pubic symphysis.[19]

Authors' Preferred Treatment

It is the authors' experience that mild symphyseal instability can be the result of disruption of the central pivot or rectus abdominis adductor aponeurosis. A repair of the central pivot appears to stabilize the pubic symphysis in this situation. For greater degrees of instability or in recalcitrant cases despite a pelvic floor repair, more aggressive surgical stabilization with open reduction internal fixation and fusion may be warranted. However, we have not encountered the need for fusion in the athletic population. In the absence of athletic pubalgia, an open or endoscopic decompression may be warranted but further outcomes are necessary to better define the results after these procedures in an athletic population.

COMPLICATIONS

Athletic Pubalgia

The most common postoperative complaint after athletic pubalgia surgery is minor bruising or edema involving the abdomen, thighs, genitals, and perineum. Postoperative hematoma requiring reoperation occurred in 0.3% of patients, and the wound infection rate was 0.4%. Nerve dysesthesia of the ilioinguinal, genitofemoral, anterior or lateral femoral cutaneous nerve distribution occurred in 0.3% of patients. Penile vein thrombosis occurred in 0.1% of patients but all resolved.[19] There is also the potential for postoperative scar tissue and subsequent neural dysesthesias. The most common reason for reoperation was development of similar symptoms on the contralateral side. The second most common was for adductor release not carried out at the first surgery.[19] Another common reason for continued disability after surgical treatment results from failure to identify associated intra-articular hip pathology (i.e., FAI).[7,19]

Osteitis Pubis

Complications associated with curettage of the symphysis for osteitis pubis include hemospermia and intermittent scrotal swelling.[30]

Adductor Tenotomy

For proximal adductor procedures there is a potential for spermatic cord injury if dissection is carried medial to the gracilis origin on the pubis.[65]

FUTURE CONSIDERATIONS

The understanding of diagnosis and treatment of athletic pubalgia is constantly improving. Although it was formerly thought to be an isolated pathology, there is now evidence to suggest otherwise. As the understanding evolves, a more athlete-specific pathology targeted approach is emerging. Based on the evidence supporting a link between athletic pubalgia and femoroacetabular impingement in athletes presenting with groin or pelvic pain, both FAI and athletic pubalgia should be considered. One important question is whether all athletes presenting with symptoms consistent with both intra-articular hip pathology and athletic pubalgia/require treatment of both entities initially. This is a challenging scenario as there is probably a transition point where addressing the FAI alone is inadequate because permanent injury has occurred to the anterior pelvic musculature, leading to athletic pubalgia. At this time there are no clear cut indicators as to when the athletic pubalgia symptoms/pathology are beyond a point where they can be relieved with restoration of hip ROM after FAI corrective procedures. When timing is an issue, both might be best treated at the same sitting.

More sensitive imaging modalities would be helpful in determining the exact location of subtle anatomic injuries for athletic pubalgia and FAI. This would give some conformity with respect to which athletic pubalgia surgical approach would be best indicated for a particular patient, since there are several variants. With future studies focusing on the variable anatomic pathology and specific approaches used to treat these entities, we may develop a better understanding of the best approaches for treating this challenging patient population.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Birmingham PM, Kelly BT, Jacobs R, et al. The effect of dynamic femoroacetabular impingement on pubic symphysis motion: a cadaveric study. *Am J Sports Med*. 2012;40(5):1113–1118.

Level of Evidence:

Controlled cadaveric laboratory study

Summary:

This is a cadaveric study looking a motion through the pubic symphysis in the native state and with a simulated cam lesion. The authors found that rotation was increased through the symphysis in the presence of dynamic femoroacetabular impingement caused by a cam lesion and hypothesized that this pathologic motion could contribute to the development of athletic pubalgia.

Citation:

Larson CM, Pierce BR, Giveans MR. Treatment of athletes with symptomatic intra-articular hip pathology and athletic pubalgia/sports hernia: a case series. *Arthroscopy*. 2011;27(6):768–775.

Level of Evidence:

IV, therapeutic case series

Summary:

In a retrospective review, the authors identified 31 athletes (37 hips) who presented with symptoms consisted with both femoroacetabular impingement and athletic pubalgia. Treating either entity in isolation resulted in low rate of return to sport (25% treated for athletic pubalgia alone and 50% treated for femoroacetabular impingement alone), leading the authors to treat both entities in a staged or concurrent fashion with 85% to 93% return to sports.

Citation:
Meyers WC, McKechnie A, Philippon MJ, et al. Experience with "sports hernia" spanning two decades. *Ann Surg*. 2008;248(4):656–665.

Level of Evidence:
IV, retrospective review

Summary:
The authors present their experience with 8490 patients with the diagnosis of sports hernia, 5218 of whom underwent surgical treatment. The authors noted an increase in percentage of patients opting for surgical treatment and a changing patient demographic over the last 20 years.

Citation:
Zoga AC, Kavanagh EC, Omar IM, et al. Athletic pubalgia and the "sports hernia": MR imaging findings. *Radiology*. 2008;247(3):797–807.

Level of Evidence:
IV, retrospective review

Summary:
The authors retrospectively reviewed the MRIs of 141 patients with symptoms with athletic pubalgia to determine the sensitivity and specificity of imaging findings compared with the standards of surgical or physical examination findings. MRI was 68% sensitive and 100% specific for injuries to the rectus abdominis and 86% sensitive and 89% specific for adductor tendon injury.

Citation:
Holmich P, Nyvold P, Larsen K. Continuing significant effect of physical training as treatment for overuse injury: 8- to 12-year outcome of a randomized clinical trial. *Am J Sports Med*. 2011;39(11):2447–2451.

Level of Evidence:
I, randomized controlled trial, 80% long-term follow-up

Summary:
In the initial randomized controlled trial, the authors compared an active physical therapy program, emphasizing core stability and eccentric strengthening, to a passive therapy program in the treatment of adductor related groin pain in athletes. At long-term follow-up, the active therapy group demonstrated sustained improvement in outcome scores compared with the passive therapy group.

REFERENCES

1. Swan K, Wolcott M. The athletic hernia: a systematic review. *Clin Orthop Relat Res*. 2007;455:78–87.
2. Meyers WC, Yoo E, Devon O, et al. Understanding "sports hernia" (athletic pubalgia): the anatomic and pathophysiologic basis for abdominal and groin pain in athletes. *Oper Tech Sports Med*. 2007;15(4):165–177.
3. Gilmore J. Groin pain in the soccer athlete: fact, fiction, and treatment. *Clin Sports Med*. 1998;17(4):787–793.
4. Taylor DC, Meyers WC, Moylan JA, et al. Abdominal musculature abnormalities as a cause of groin pain in athletes: Inguinal hernias and pubalgia. *Am J Sports Med*. 1991;19(3): 239–242.
5. Verrall GM, Slavotinek JP, Fon GT, et al. Outcome of conservative management of athletic chronic groin injury diagnosed as pubic bone stress injury. *Am J Sports Med*. 2007;35(3):467–474.
6. Meyers WC, Foley DP, Garrett WE, et al. Management of severe lower abdominal or inguinal pain in high-performance athletes. PAIN (Performing Athletes with Abdominal or Inguinal Neuromuscular Pain Study Group). *Am J Sports Med*. 2000;28(1):2–8.
7. Larson CM, Pierce BR, Giveans MR. Treatment of athletes with symptomatic intra-articular hip pathology and athletic pubalgia/ sports hernia: a case series. *Arthroscopy*. 2011;27(6):768–775.
8. Brunt LM. Hernia management in athlete. *Adv Surg*. 2016;50(1):187–202.
9. Gamble JG, Simmons SC, Freedman M. The symphysis pubis. Anatomic and pathologic considerations. *Clin Orthop Relat Res*. 1986;203:261–272.
10. Feeley BT, Powell JW, Muller MS, et al. Hip injuries and labral tears in the National Football League. *Am J Sports Med*. 2008;36(11):2187–2195.
11. Mullens FE, Zoga AC, Morrison WB, et al. Review of MRI technique and imaging findings in athletic pubalgia and the "sports hernia". *Eur J Radiol*. 2011 Sep 3; [Epub ahead of print].
12. Birmingham P, Larson CM. Medial soft tissue injuries of the hip: adductor strains and athletic pubalgia. In: Kelly BT, Larson CM, Bedi A, eds. *Sports Hip Injuries: Diagnosis and Management*. New Jersey: SLACK Inc; 2012. In publication.
13. Minnich JM, Hanks JB, Muschaweck U, et al. Sports hernia: diagnosis and treatment highlighting a minimal repair surgical technique. *Am J Sports Med*. 2011;39(6):1341–1349.
14. Verrall GM, Hamilton IA, Slavotinek JP, et al. Hip joint range of motion reduction in sports-related chronic groin injury diagnosed as pubic bone stress injury. *J Sci Med Sport*. 2005;8(1):77–84.
15. Birmingham PM, Kelly BT, Jacobs R, et al. The effect of dynamic femoroacetabular impingement on pubic symphysis motion: a cadaveric study. *Am J Sports Med*. 2012;40(5):1113–1118.
16. Spinelli A. A new sports injury: pubic pain in fencers. *Orthop Traumatol dell'Apparato Motore*. 1932;4:111–127.
17. Gilmore OJ. Gilmore's groin: ten years experience of groin disruption–a previously unsolved problem in sportsmen. *Sports Med Soft Tissue Trauma*. 1991;1(3):12–14.
18. Hackney RG. The sports hernia: a cause of chronic groin pain. *Br J Sports Med*. 1993;27(1):58–62.
19. Irshad K, Feldman LS, Lavoie C, et al. Operative management of "hockey groin syndrome": 12 years experience in National Hockey League players. *Surgery*. 2001;130(4):759–764, discussion 764–766.
20. Meyers WC, McKechnie A, Philippon MJ, et al. Experience with "sports hernia" spanning two decades. *Ann Surg*. 2008;248(4): 656–665.
21. Knapik DM, Gebhart JJ, Nho SJ, et al. Prevalence of surgical repair for athletic pubalgia and impact on performance in football athletes participating in the National Football League combine. *Arthroscopy*. 2017;33(5):1044–1049. doi:10.1016/j. arthro.2017.01.024. [Epub 2017 Mar 9].
22. Freely BT, Powell JW, Muller MS, et al. Hip injuries and labral tears in the National Football League. *Am J Sports Med*. 2008;36(11):2187–2195. doi:10.1177/0363546508319898. [Epub 2008 Jul 18].
23. Deleted in review.
24. Deleted in review.
25. Deleted in review.
26. Deleted in review.
27. Deleted in review.
28. Farber AJ, Wilckens JH. Sports hernia: diagnostic and therapeutic approach. *J Am Acad Orthop Surg*. 2007;15(8): 507–514.
29. Williams PR, Thomas DP, Downes EM. Osteitis pubis and instability of the pubic symphysis. When nonoperative measures fail. *Am J Sports Med*. 2000;28(3):350–355.
30. Williams PR, Thomas DP, Downes EM. Osteitis pubis and instability of the pubic symphysis. When nonoperative measures fail. *Am J Sports Med*. 2000;28(3):350–355.
31. Deleted in review.
32. Clohisy JC, Carlisle JC, Beaule PE, et al. A systematic approach to the plain radiographic evaluation of the young adult hip. *J Bone Joint Surg Am*. 2008;90(suppl 4):47–66.
33. Zoga AC, Kavanagh EC, Omar IM, et al. Athletic pubalgia and the "sports hernia": MR imaging findings. *Radiology*. 2008; 247(3):797–807.
34. Slavotinek JP, Verrall GM, Fon GT, et al. Groin pain in footballers: the association between preseason clinical and pubic bone magnetic resonance imaging findings and athlete outcome. *Am J Sports Med*. 2005;33(6):894–899.
35. Muschaweck U, Berger L. Sportsmen's groin-diagnostic approach and treatment with the minimal repair technique: a single-center uncontrolled clinical review. *Sports Health*. 2010;2(3):216–221.
36. Tyler TF, Nicholas SJ, Campbell RJ, et al. The association of hip strength and flexibility with the incidence of adductor muscle strains in professional ice hockey players. *Am J Sports Med*. 2001;29(2):124–128.
37. Tyler TF, Nicholas SJ, Campbell RJ, et al. The effectiveness of a preseason exercise program to prevent adductor muscle strains in professional hockey players. *Am J Sports Med*. 2002;30(5): 680–683.
38. Holmich P, Uhrskou P, Ulnits L, et al. Effectiveness of active physical training as a treatment for long-standing adductor related groin pain in athletes: a randomized clinical trial. *Lancet*. 1999;353(9151):439–443.
39. Holmich P, Nyvold P, Larsen K. Continuing significant effect of physical training as treatment for overuse injury: 8- to 12-year outcome of a randomized clinical trial. *Am J Sports Med*. 2011;39(11):2447–2451.
40. Singh JR, Roza R, Bartolozzi AR. Platelet rich plasma therapy in an athlete with adductor longus tendon tear. *Univ of Pennsylvania Orthop J*. 2010;20:42–43.
41. Dallaudiere B, Pesquer L, Meyer P, et al. Intratendinous injection of platelet-rich plasma under US guidance to treat tendinopathy:

a long-term pilot study. *J Vasc Interv Radiol*. 2014;25(5):717–723. doi:10.1016/j.jvir.2014.01.026. [Epub 2014 Mar 20].
42. Schlegel TF, Bushnell BD, Godfrey J, et al. Success of nonoperative management of adductor longus tendon ruptures in National Football League athletes. *Am J Sports Med*. 2009;37(7):1394–1399.
43. Akermark C, Johansson C. Tenotomy of the adductor longus tendon in the treatment of chronic groin pain in athletes. *Am J Sports Med*. 1992;20(6):640–643.
44. Atkinson HD, Johal P, Falworth MS, et al. Adductor tenotomy: its role in the management of sports related chronic groin pain. *Arch Orthop Trauma Surg*. 2010;130(8):965–970.
45. Van Der Donckt K, Steenbrugge F, Van Den Abbeele K, et al. Bassini's hernial repair and adductor longus tenotomy in the treatment of chronic groin pain in athletes. *Acta Orthop Belg*. 2003;69(1):35–41.
46. Deleted in review.
47. Gentisaris M, Goulimaris I, Sikas N. Laparoscopic repair of groin pain in athletes. *Am J Sports Med*. 2004;32(5):1238–1242.
48. Kluin J, den Hoed PT, van Linschoten R, et al. Endoscopic evaluation and treatment of groin pain in the athlete. *Am J Sports Med*. 2004;32(4):944–949.
49. Ingoldby CJ. Laparascopic and conventional repair of groin disruption in sportsmen. *Br J Surg*. 1997;84(2):213–215.
50. Muschaweck U, Berger L. Minimal repair technique of sportsmen's groin: an innovative open suture repair technique to treat chronic inguinal pain. *Hernia*. 2010;14(1):27–33.
51. Muschaweck U, Berger L. Sportsmen's groin-diagnostic approach and treatment with the minimal repair technique: a single-center uncontrolled clinical review. *Sports Health*. 2010;2(3):216–221.
52. Paajanen H, Brinck T, Hernumen H, et al. Laparoscopic surgery for chronic groin pain in athletes is more effective than nonoperative treatment: a randomized clinical trial with magnetic resonance imaging of 60 patients with sportsman's hernia (athletic pubalgia). *Surgery*. 2011;150(1):99–107.
53. Choi H, McCartney M, Best TM. Treatment of osteitis pubis and osteomyelitis of the pubic symphysis in athletes: a systematic review. *Br J Sports Med*. 2011;45(1):57–64.
54. Rodriguez C, Miguel A, Lima H, et al. Osteitis pubis syndrome in the professional soccer athlete: a case report. *J Athl Train*. 2001;36(4):437–440.
55. O'Connell MJ, Powell T, McCaffrey NM, et al. Symphyseal cleft injection in the diagnosis and treatment of osteitis pubis in athletes. *AJR Am J Roentgenol*. 2002;179(4):955–959.
56. Holt MA, Keene JS, Graf BK, et al. Treatment of osteitis pubis in athletes. Results of corticosteroid injections. *Am J Sports Med*. 1995;23(5):601–606.
57. Radic R, Annear P. Use of pubic symphysis curettage for treatment-resistant osteitis pubis in athletes. *Am J Sports Med*. 2008;36(1):122–128.
58. Grace JN, Sim FH, Shives TC, et al. Wedge resection of the symphysis pubis for the treatment of osteitis pubis. *J Bone Joint Surg Am*. 1989;71(3):358–364.
59. Schnute WJ. Osteitis pubis. *Clin Orthop*. 1961;20:187–192.
60. Moore RS Jr, Stover MD, Matta JM. Late posterior instability of the pelvis after resection of the pubic symphysis for treatment of osteitis pubis. A report of two cases. *J Bone Joint Surg Am*. 1998;80(7):1043–1048.
61. Paajanen H, Heikkinen J, Hermunen H, et al. Successful treatment of osteitis pubis by using totally extraperitoneal endoscopic technique. *Int J Sports Med*. 2005;26(4):303–306.
62. Paajanen H, Hermunen H, Karonen J. Pubic magnetic resonance imaging findings in surgically and conservatively treated athletes with osteitis pubis compared to asymptomatic athletes during heavy training. *Am J Sports Med*. 2008;36(1):117–121.
63. Matsuda DK. Endoscopic pubic symphysectomy for recalcitrant osteitis pubis associated with bilateral femoroacetabular impingement. *Orthopedics*. 2010;199–203.
64. Matsuda DK, Ribas M, Matsuda NA, et al. Multicenter outcomes of endoscopic pubic symphysectomy for osteitis pubis associated with femoroacetabular impingement. *Arthroscopy*. 2015;31(7):1255–1260. doi:10.1016/j.arthro.2015.02.005. [Epub 2015 Mar 29].
65. Larson CM. Arthroscopic management of pincer-type impingement. *Sports Med Arthrosc*. 2010;18(2):100–107.